

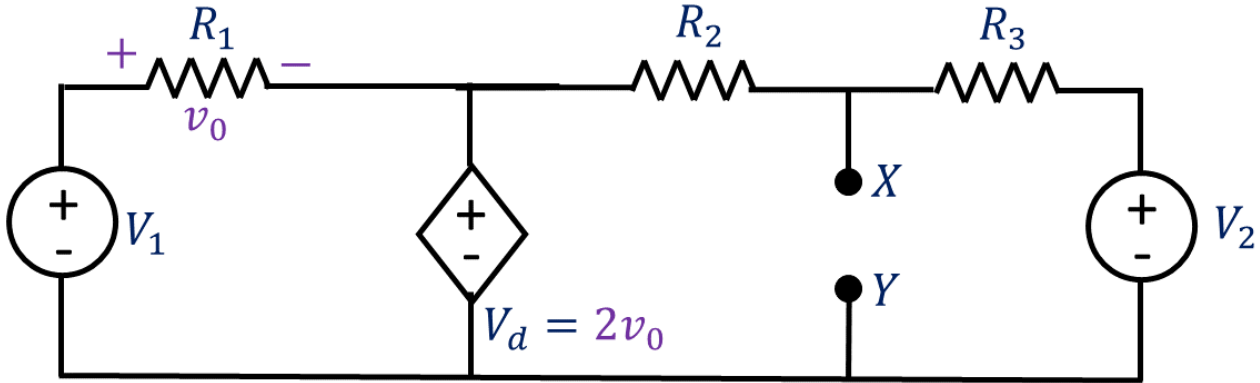


Figure 1 The network shown in the above figure has one dependent voltage source $V_d = 2v_0$, two independent voltage sources $V_1 = 18V$ and $V_2 = 15V$ along with three resistors $R_1 = 1\Omega$, $R_2 = 5\Omega$ and $R_3 = 4\Omega$. A load resistance can be connected across the two accessible terminals X and Y .

1. Consider the network shown in Figure 1. Estimate the numerical value of the optimal resistive load R_L^* and the maximum power P_{max} that will be drawn by R_L^* from this network.

Solution

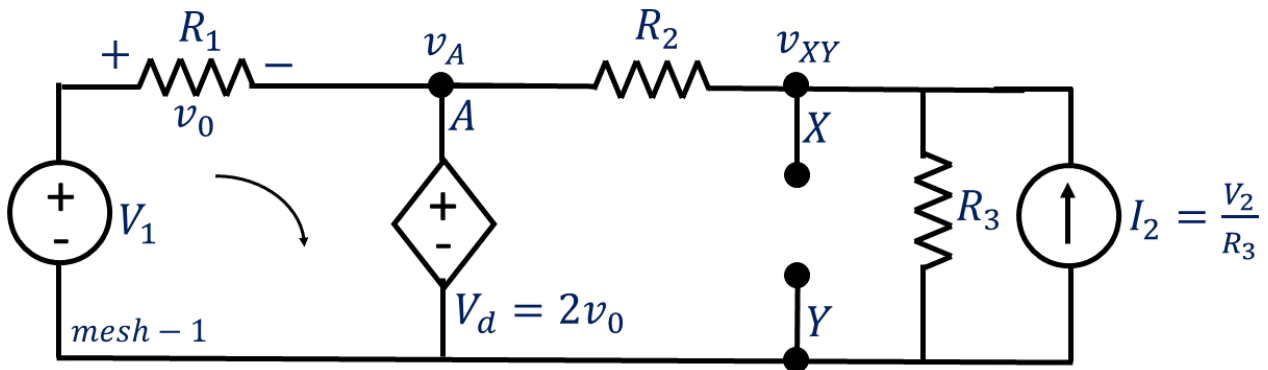


Figure 2 The voltage source V_2 in series with R_3 (Figure 1) is converted to a current source $I_2 = \frac{V_2}{R_3}$ with R_3 in parallel (source transformation). The nodes A , X and Y are marked. Here, Y is considered as reference node.

KVL in mesh-1 (Figure 2) gives

$$-V_1 + v_0 + 2v_0 = 0 \quad \Rightarrow \quad v_0 = \frac{V_1}{3}$$

Using numerical values we get

$$v_A = 2v_0 = 2 \times \frac{V_1}{3} = 2 \times \frac{18}{3} V = 12V$$

KCL at node X (Figure 2) gives,

$$\begin{aligned} & \frac{v_A - v_{XY}}{R_2} - \frac{v_{XY}}{R_3} + \frac{V_2}{R_3} = 0 \\ \Rightarrow & v_{XY} \left(\frac{1}{R_2} + \frac{1}{R_3} \right) = \frac{v_A}{R_2} + \frac{V_2}{R_3} \\ \Rightarrow & v_{XY} = \frac{v_A R_3 + R_2 V_2}{R_2 + R_3} \end{aligned}$$

Using numerical values we get

$$v_{XY} = \frac{12 \times 4 + 5 \times 15}{4 + 5} V = 13.67V$$

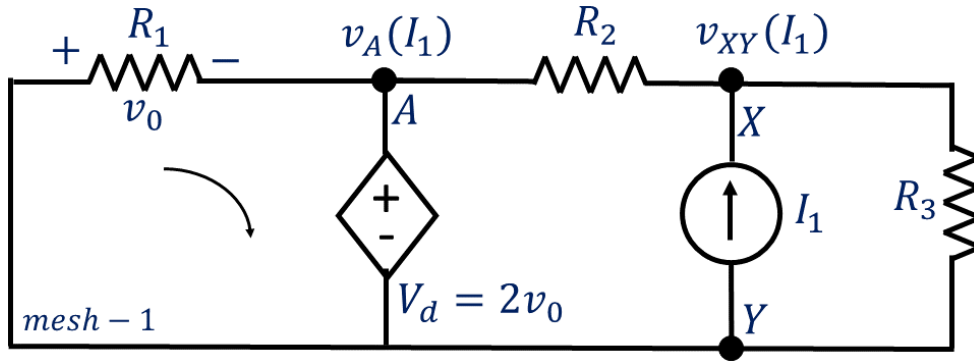


Figure 3 The independent voltage sources in the network are deactivated (or short-circuited). An independent current source $I_1 = 1A$ is connected between terminals X and Y as a test source for measuring the equivalent resistance R_{TH} of the network. Here, Node Y is taken as reference.

KVL in mesh-1 (Figure 3) gives

$$v_0 + 2v_0 = 0 \quad \Rightarrow v_0 = 0 \quad \Rightarrow V_A(I_1) = 2v_0 = 0$$

KCL at node X gives (Figure 3)

$$\begin{aligned} & \frac{v_{XY}(I_1) - v_A(I_1)}{R_2} + \frac{v_{XY}(I_1)}{R_3} = I_1 \\ \Rightarrow & v_{XY}(I_1) \left(\frac{1}{R_2} + \frac{1}{R_3} \right) = 1 \quad (\text{because } v_A(I_1) = 0 \text{ and } I_1 = 1A) \\ \Rightarrow & v_{XY}(I_1) = \frac{R_2 R_3}{R_2 + R_3} \end{aligned}$$

Using numerical values we get,

$$v_{XY}(I_1) = \frac{4 \times 5}{4 + 5} V = 2.22V$$

Thus,

$$R_{TH} = \frac{v_{XY}(I_1)}{I_1} = 2.22\Omega$$

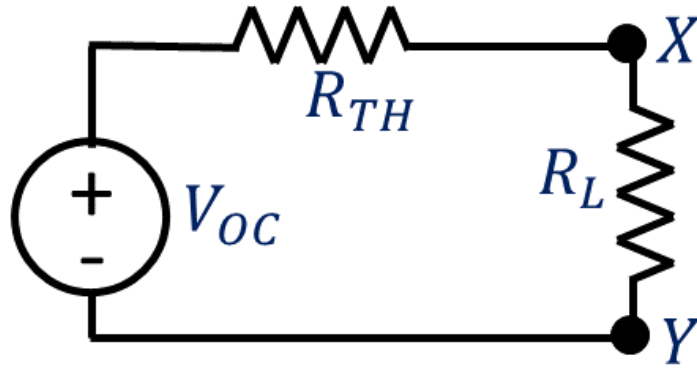


Figure 4 The Thevenin's equivalent circuit for the network of Figure 1, as seen from terminals $X - Y$. Here, $V_{OC} = 13.67V$ and $R_{TH} = 2.22\Omega$.

According to the maximum power transfer theorem, the network will deliver the maximum power P_{max} to the load of optimal resistance R_L^* when,

$$R_L^* = R_{TH} = 2.22\Omega$$

$$\text{and, } P_{max} = \frac{V_{OC}^2}{4R_{TH}} = \frac{(13.67)^2}{4 \times 2.22} W = 21.044W$$

2. We need to design a code converter from the Excess-3 code to the 2-out-of-5 code shown in the figure below. Find minimal SOP expressions and minimal POS expressions for each output.

Decimal	2-out-of-5				
0	1	1	0	0	0
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	0
7	1	0	0	0	1
8	1	0	0	1	0
9	1	0	1	0	0

Solution

Let the Excess-3 input be represented by four bits $ABCD$ and the 2-out-of-5 output be represented by five bits $PQRST$. The truth table relating the inputs to the outputs can be seen as below:

Decimal	A	B	C	D	P	Q	R	S	T
0	0	0	1	1	1	1	0	0	0
1	0	1	0	0	0	0	0	1	1
2	0	1	0	1	0	0	1	0	1
3	0	1	1	0	0	0	1	1	0
4	0	1	1	1	0	1	0	0	1
5	1	0	0	0	0	1	0	1	0
6	1	0	0	1	0	1	1	0	0
7	1	0	1	0	1	0	0	0	1
8	1	0	1	1	1	0	0	1	0
9	1	1	0	0	1	0	1	0	0

The minimal SOP expressions can be obtained as follows from the Karnaugh Maps for P , Q , R , S and T .

AB \ CD	00	01	11	10
00	ϕ	ϕ	1	ϕ
01	0	0	0	0
11	1	ϕ	ϕ	ϕ
10	0	0	1	1

(a) K-Map for P

AB \ CD	00	01	11	10
00	ϕ	ϕ	1	ϕ
01	0	0	1	0
11	0	ϕ	ϕ	ϕ
10	1	1	0	0

(b) K-Map for Q

AB \ CD	00	01	11	10
00	ϕ	ϕ	0	ϕ
01	0	1	0	1
11	1	ϕ	ϕ	ϕ
10	0	1	0	0

(c) K-Map for R

AB \ CD	00	01	11	10
00	ϕ	ϕ	0	ϕ
01	1	0	0	1
11	0	ϕ	ϕ	ϕ
10	1	0	1	0

(d) K-Map for S

AB \ CD	00	01	11	10
00	ϕ	ϕ	0	ϕ
01	1	1	1	0
11	0	ϕ	ϕ	ϕ
10	0	0	0	1

(e) K-Map for T

$$P = A.B + \overline{B}.C$$

$$Q = \overline{B}.\overline{C} + \overline{A}.C.D$$

$$R = A.B + \overline{C}.D + \overline{A}.C.\overline{D} \text{ or } A.B + \overline{C}.D + B.C.\overline{D}$$

$$S = \overline{A}.\overline{D} + A.C.D + \overline{B}.\overline{C}.\overline{D}$$

$$T = \overline{A}.\overline{C} + B.D + A.C.\overline{D} \text{ or } \overline{A}.\overline{C} + B.D + \overline{B}.C.\overline{D}$$

The minimal POS expressions can be obtained as follows from the Karnaugh Maps for P , Q , R , S and T .

AB \ CD	00	01	11	10
00	ϕ	ϕ	1	ϕ
01	0	0	0	0
11	1	ϕ	ϕ	ϕ
10	0	0	1	1

(a) K-Map for P

AB \ CD	00	01	11	10
00	ϕ	ϕ	1	ϕ
01	0	0	1	0
11	0	ϕ	ϕ	ϕ
10	1	1	0	0

(b) K-Map for Q

AB \ CD	00	01	11	10
00	ϕ	ϕ	0	ϕ
01	0	1	0	1
11	1	ϕ	ϕ	ϕ
10	0	1	0	0

(c) K-Map for R

AB \ CD	00	01	11	10
00	ϕ	ϕ	0	ϕ
01	1	0	0	1
11	0	ϕ	ϕ	ϕ
10	1	0	1	0

(d) K-Map for S

AB \ CD	00	01	11	10
00	ϕ	ϕ	0	ϕ
01	1	1	1	0
11	0	ϕ	ϕ	ϕ
10	0	0	0	1

(e) K-Map for T

$$P = (A + \overline{B}).(B + C)$$

$$Q = (\overline{A} + \overline{C}).(\overline{B} + C).(\overline{B} + D) \quad \text{or} \quad (\overline{A} + \overline{C}).(\overline{B} + C).(A + D) \quad \text{or} \quad (\overline{A} + \overline{C}).(\overline{B} + C).(\overline{C} + D) \\ \text{or} \quad (\overline{A} + \overline{C}).(A + C).(\overline{B} + D)$$

$$R = (\overline{C} + \overline{D}).(A + C + D).(B + D)$$

$$S = (\overline{A} + \overline{B}).(C + \overline{D}).(A + \overline{D}).(\overline{A} + \overline{C} + D) \quad \text{or} \quad (\overline{A} + \overline{B}).(C + \overline{D}).(A + \overline{D}).(B + \overline{C} + D)$$

$$T = (\overline{A} + C).(B + \overline{D}).(A + \overline{C} + D) \quad \text{or} \quad (\overline{A} + C).(B + \overline{D}).(\overline{B} + \overline{C} + D)$$

3. We need to design a code converter from the BCD code to the Ringtail code shown in the figure below. Find minimal SOP expressions and minimal POS expressions for each output.

Decimal	Ringtail				
0	0	0	0	0	0
1	0	0	0	0	1
2	0	0	0	1	1
3	0	0	1	1	1
4	0	1	1	1	1
5	1	1	1	1	1
6	1	1	1	1	0
7	1	1	1	0	0
8	1	1	0	0	0
9	1	0	0	0	0

Solution

Let the BCD input be represented by four bits $ABCD$ and the Ringtail output be represented by five bits $PQRST$. The truth table relating the inputs to the outputs can be seen as below:

Decimal	A	B	C	D	P	Q	R	S	T
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1
2	0	0	1	0	0	0	0	1	1
3	0	0	1	1	0	0	1	1	1
4	0	1	0	0	0	1	1	1	1
5	0	1	0	1	1	1	1	1	1
6	0	1	1	0	1	1	1	1	0
7	0	1	1	1	1	1	1	0	0
8	1	0	0	0	1	1	0	0	0
9	1	0	0	1	1	0	0	0	0

CD \ AB	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	ϕ	ϕ	ϕ	ϕ
10	1	1	ϕ	ϕ

(a) K-Map for P

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	ϕ	ϕ	ϕ	ϕ
10	1	0	ϕ	ϕ

(b) K-Map for Q

CD \ AB	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	ϕ	ϕ	ϕ	ϕ
10	0	0	ϕ	ϕ

(c) K-Map for R

CD \ AB	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	ϕ	ϕ	ϕ	ϕ
10	0	0	ϕ	ϕ

(d) K-Map for S

CD \ AB	00	01	11	10
00	0	1	1	1
01	1	1	0	0
11	ϕ	ϕ	ϕ	ϕ
10	0	0	ϕ	ϕ

(e) K-Map for T

The minimal SOP expressions can be obtained as follows from the Karnaugh Maps for P , Q , R , S and T .

$$P = A + B.C + B.D$$

$$Q = B + A.\bar{D}$$

$$R = B + C.D$$

$$S = \bar{B}.C + B.\bar{C} + \textcolor{blue}{C}.\bar{D} \text{ or } \bar{B}.C + B.\bar{C} + \textcolor{red}{B}.\bar{D}$$

$$T = \bar{B}.C + B.\bar{C} + \textcolor{red}{A}.\bar{B}.D \text{ or } \bar{B}.C + B.\bar{C} + \textcolor{blue}{A}.\bar{C}.D$$

CD \ AB	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	ϕ	ϕ	ϕ	ϕ
10	1	1	ϕ	ϕ

(a) K-Map for P

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	ϕ	ϕ	ϕ	ϕ
10	1	0	ϕ	ϕ

(b) K-Map for Q

CD \ AB	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	ϕ	ϕ	ϕ	ϕ
10	0	0	ϕ	ϕ

(c) K-Map for R

CD \ AB	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	ϕ	ϕ	ϕ	ϕ
10	0	0	ϕ	ϕ

(d) K-Map for S

CD \ AB	00	01	11	10
00	0	1	1	1
01	1	1	0	0
11	ϕ	ϕ	ϕ	ϕ
10	0	0	ϕ	ϕ

(e) K-Map for T

The minimal POS expressions can be obtained as follows from the Karnaugh Maps for P , Q , R , S and T .

$$P = (A + B).(A + C + D)$$

$$Q = (A + B).(B + \overline{D}) \quad \text{or} \quad (A + B).(\overline{A} + \overline{D})$$

$$R = (B + C).(B + D)$$

$$S = (B + C).(\overline{B} + \overline{C} + \overline{D})$$

$$T = (\overline{A}).(\overline{B} + \overline{C}).(B + C + D)$$